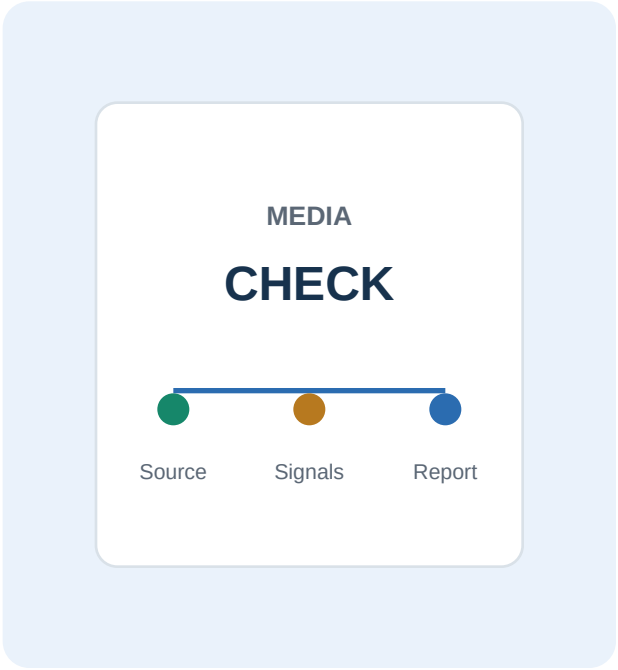
SCANLINE

Verify media.
Understand the result.
Respond with confidence.

An explainable deepfake detection and support system for safer decisions around digital media.



Project presentation | Prepared by Pawan Kumar, Abhijeet Kumar Thakur, Suryan Karan Roshan and Divya Raj

WHY SCANLINE

The problem: trust is now a decision, not a default.

Synthetic and manipulated media can travel quickly. People need more than a binary label: they need a clear result, a reason to review it, and a safe next action.

Misinformation

Misleading images can distort public understanding before verification catches up.

Fraud and impersonation

Altered media can be used to pressure, deceive or imitate a trusted person.

Evidence uncertainty

Investigators and users need a repeatable way to preserve and review media.

THE SCANLINE PROPOSITION

A practical system: detect, explain, protect.

1 Detect

Analyze uploaded media for multiple manipulation signals. Present a calibrated confidence range, not certainty.

2 Explain

Show the strongest contributing signals in plain language and provide an inspection view for reviewers.

3 Protect

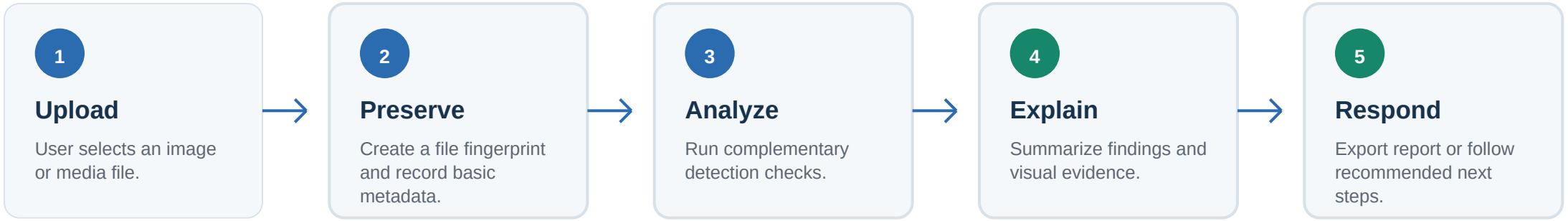
Create a shareable record, preserve the original file details, and guide the user to appropriate next steps.

Design principle

ScanLine supports human judgement. It is a decision aid, not a final authority on authenticity.

END-TO-END WORKFLOW

How ScanLine works



Each stage produces a traceable record so a reviewer can understand what was assessed and what should happen next.

SIGNAL CATEGORIES

What the system examines

Visual consistency

Lighting, shadows, texture, edges and local image artifacts that may warrant closer review.

Image provenance

Available metadata, file structure and duplicate or altered-file indicators.

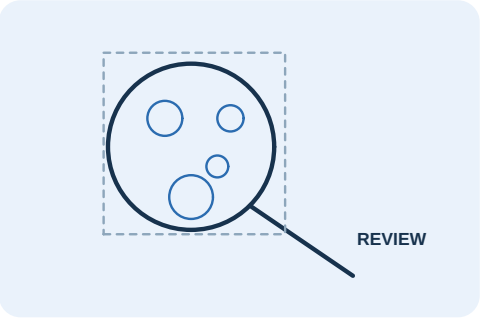
Model-based cues

Patterns identified by trained classifiers and explainability overlays.

No single signal proves that a file is real or fake. Combining independent checks makes the review more useful and more transparent.

Important

A result should be treated as a screening outcome and escalated for human review when the impact is high.



USER EXPERIENCE

From upload to an understandable result

- **Simple input**

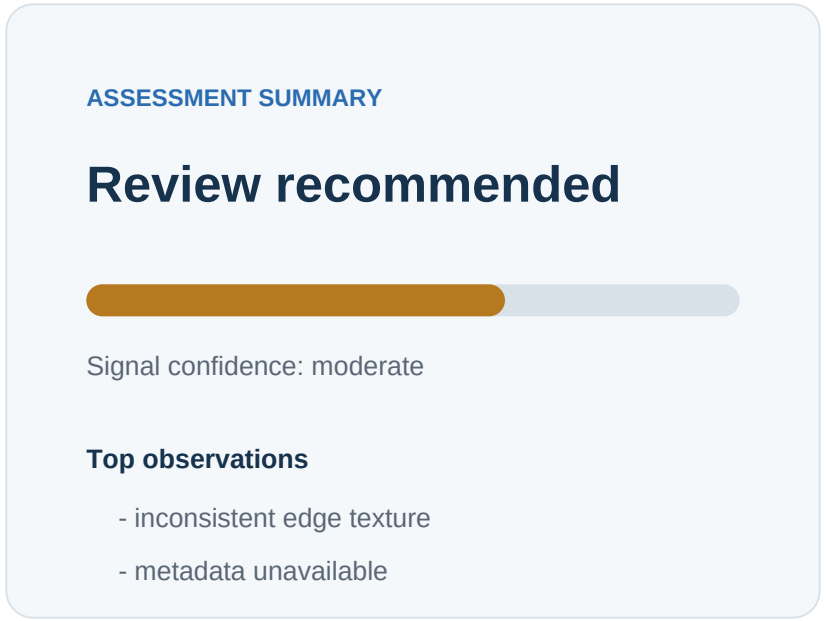
  A focused upload screen accepts a file and explains what data is retained for the assessment.

- **Clear result**

  A review status, confidence range and concise summary help users interpret the outcome without technical jargon.

- **Useful actions**

  Users can save a report, retain the original, seek a second review or follow support guidance.



PRODUCT SCOPE

Core capabilities in one place

Single and batch checks

Review one file or organize a controlled set for triage.

Review history

Keep a searchable log of submitted assessments and reports.

Confidence communication

Use plain-language statuses alongside technical context.

Evidence package

Export a dated summary with original-file reference information.

Explainability view

Surface the signals that contributed most to the model output.

Guided escalation

Direct users to internal, legal or cybercrime support channels as appropriate.

RESPONSIBLE AI

Explainability turns a score into a conversation.

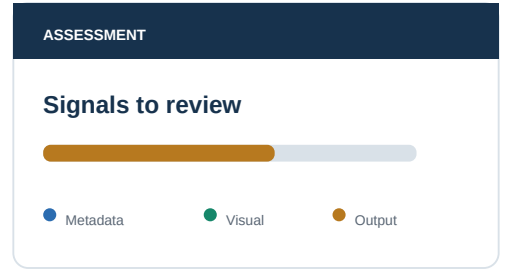
A useful system shows why a result deserves attention, while making its uncertainty explicit. ScanLine can pair a model output with a visual heatmap and a short explanation of the strongest observed signals.

What users see

Status - confidence range - top signals - visual inspection area - recommended next step

What reviewers see

File reference - processing time - model version - reviewed signals - exportable notes



Outcome

The result is easier to challenge, compare and document - essential when decisions affect people.

TRUST AND SAFETY

Preserve evidence without overclaiming.

- **Record the original**

  Retain the submitted file reference, file hash, upload time and any available metadata.

- **Separate assessment from verdict**

  Clearly distinguish model signals from factual conclusions or legal determinations.

- **Protect sensitive material**

  Use access controls, defined retention periods and secure sharing for reports and source media.

REPORT RECORD

- File fingerprint
- Assessment time
- Model and version
- Reviewer notes

IMPLEMENTATION ROADMAP

A staged path from prototype to dependable service

Phase 1	**Prototype**	Image intake, baseline detection, simple report.
Phase 2	**Review workflow**	Explainability screen, history, role-based review.
Phase 3	**Quality and security**	Validation set, monitoring, privacy and retention controls.
Phase 4	**Broader access**	Browser workflow, multilingual guidance and integration options.

Each phase should be evaluated against real user needs, safety requirements and documented performance limits.

VALIDATION AND GOVERNANCE

Measure what matters, then communicate it honestly.

Quality checks

Measure performance across varied sources, compression levels, demographics and manipulation types.

Human review

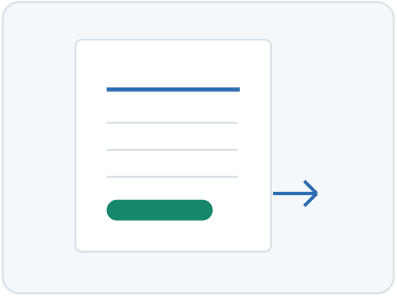
Test whether people can understand the result, identify uncertainty and take the right next action.

Ongoing monitoring

Track drift, false positives, false negatives and feedback from reviewers after release.

Recommended safeguards

Versioned models - clear limitations - escalation path - audit logs - periodic independent review



Detect.
Explain.
Protect.

ScanLine is designed to make deepfake assessments clearer, more traceable and more actionable - with people kept in the loop.

THANK YOU

Questions?